WSQ · PROCESS INTEGRATION

Mastering Semiconductor Key Modules for Enhanced CMOS Process Integration and Device Performance

Connect modules · control interactions · verify device performance

TGS-2024051248

Version v9.0 · 20 August 2026

Trainer: Dr Alfred Ang

Tertiary Infotech Academy Pte Ltd · UEN 201200696W



Digital Attendance (Mandatory)

1. Take AM, PM and Assessment digital attendance for this WSQ course.
2. The trainer or administrator displays the SSG attendance QR code.
3. Scan with Singpass/mobile camera and submit each required attendance.
4. At least 75% attendance and a Competent outcome are required.

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About the Trainer - General Template



Your Trainer

Complete before delivery

NAME

_____

TITLE / DESIGNATION

_____

QUALIFICATIONS

_____

AREAS OF EXPERTISE

_____

TRAINING EXPERIENCE

_____

CONTACT

_____

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TRAINER PROFILE

About the Trainer - Dr Alfred Ang



Dr Alfred Ang

Principal Trainer

ROLE
Principal Trainer

QUALIFICATIONS
Doctorate · ACTA/ACLP-aligned adult educator

EXPERTISE
Semiconductor · quality · emerging technology

EXPERIENCE
Professional training and technology leadership

CONTACT
enquiry@tertiaryinfotech.com

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ICEBREAKER

Let's Know Each Other

1. Name, organisation and role
2. Your semiconductor process or quality experience
3. One module interaction you find difficult
4. One result you want by the end of Day 2

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HOUSEKEEPING

Ground Rules

1. **Evidence first**

distinguish measurements from assumptions

2. **Participate actively and ask technical questions**

Participate actively and ask technical questions

3. **Respect confidential process and product information**

Respect confidential process and product information

4. **One conversation at a time**

One conversation at a time

5. **Return from breaks punctually**

Return from breaks punctually

6. **Stop any unsafe or unauthorised activity**

Stop any unsafe or unauthorised activity

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COURSE AT A GLANCE

Course Profile

Programme

1. 2 days · 16 hours
2. TSC: Process Integration
3. ELE-SIS-4002-1.1
4. Official Topic 1: process flow

Evidence and assessment

1. Official Topic 2: integration and performance
2. Eight individual activity folders
3. WA-SAQ and Practical Performance
4. Open-book assessment on Day 2

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1

Identify process flows and modules for specific CMOS technologies in manufacturing.

2

LO2

Determine CMOS device performance and verify process integration in accordance with product specifications.

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Knowledge Coverage

1

K1

Specific technology process flow

2

K2

Types of performance metrics

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ABILITIES A1-A5

Ability Coverage

1

A1

Identify process flow for the specific technology in the manufacturing process in accordance with technology specifications

2

A2

Determine process interaction between and within process modules to the specific technology in accordance with technology specifications

3

A3

Determine device performance in accordance with product specifications

4

A4

Verify process integration performance to the specific technology in accordance with product specifications

5

A5

Determine follow-up actions required

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SCHEDULE

Lesson Plan - Day 1

Morning

1. 09:00 Admin, outcomes and assessment briefing
2. 09:20 Block 1A: patterning and etch
3. 10:40 Activities 1-2
4. 12:30 Lunch

Afternoon

1. 13:15 Block 1B: implant, thermal and films
2. 15:05 Activities 3-4
3. 17:20 Integrated-flow review
4. 18:00 End of Day 1

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SCHEDULE

Lesson Plan - Day 2

Morning

1. 09:00 Block 2A: performance, yield and reliability
2. 10:45 Activities 5-6
3. 12:30 Lunch
4. 13:15 Block 2B: SPC and packaging

Afternoon and assessment

1. 14:15 Activities 7-8 and course synthesis
2. 16:00 Written Assessment · 60 min
3. 17:00 Practical Performance · 60 min
4. 18:00 End of class

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BEFORE ASSESSMENT

Briefing for Assessment

1. Individual, open-book work; no discussion
2. Use only the slides, Learner Guide and approved materials
3. No photographs or recordings of assessment scripts
4. Submit on the LMS and sign the Assessment Summary Record

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ASSESSMENT

Final Assessment

Knowledge

1. Written Assessment (WA-SAQ)
2. 2 open-ended questions
3. K1-K2
4. 60 minutes

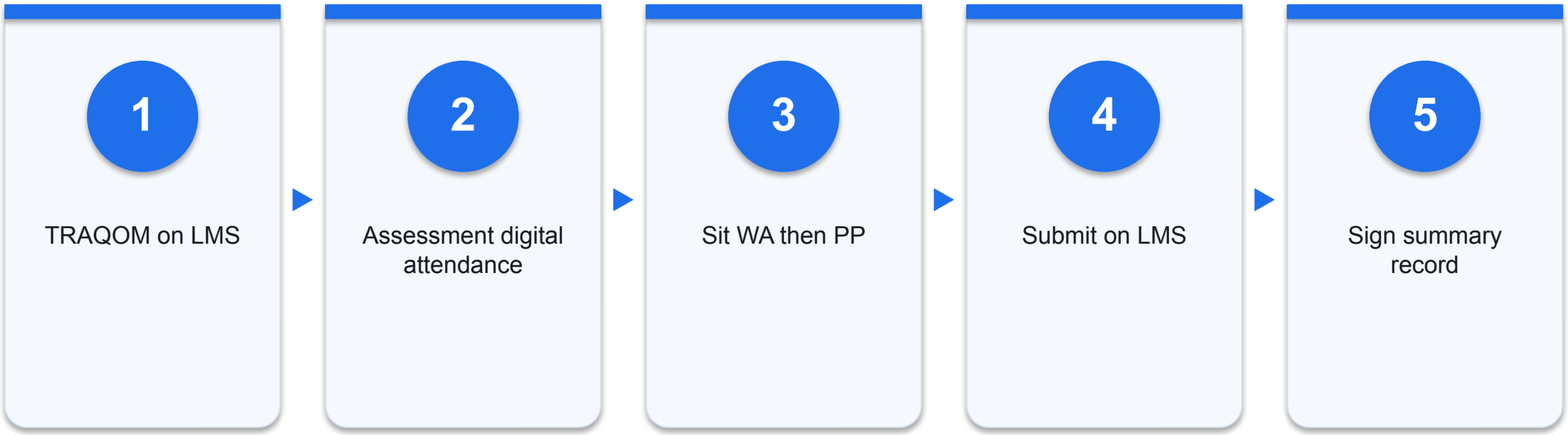
Ability

1. Practical Performance
2. 2 integrated tasks
3. A1-A5
4. 60 minutes

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ON ASSESSMENT DAY

Assessment Flow



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Access the Course Activities



Open the course repository to access all eight activity folders.

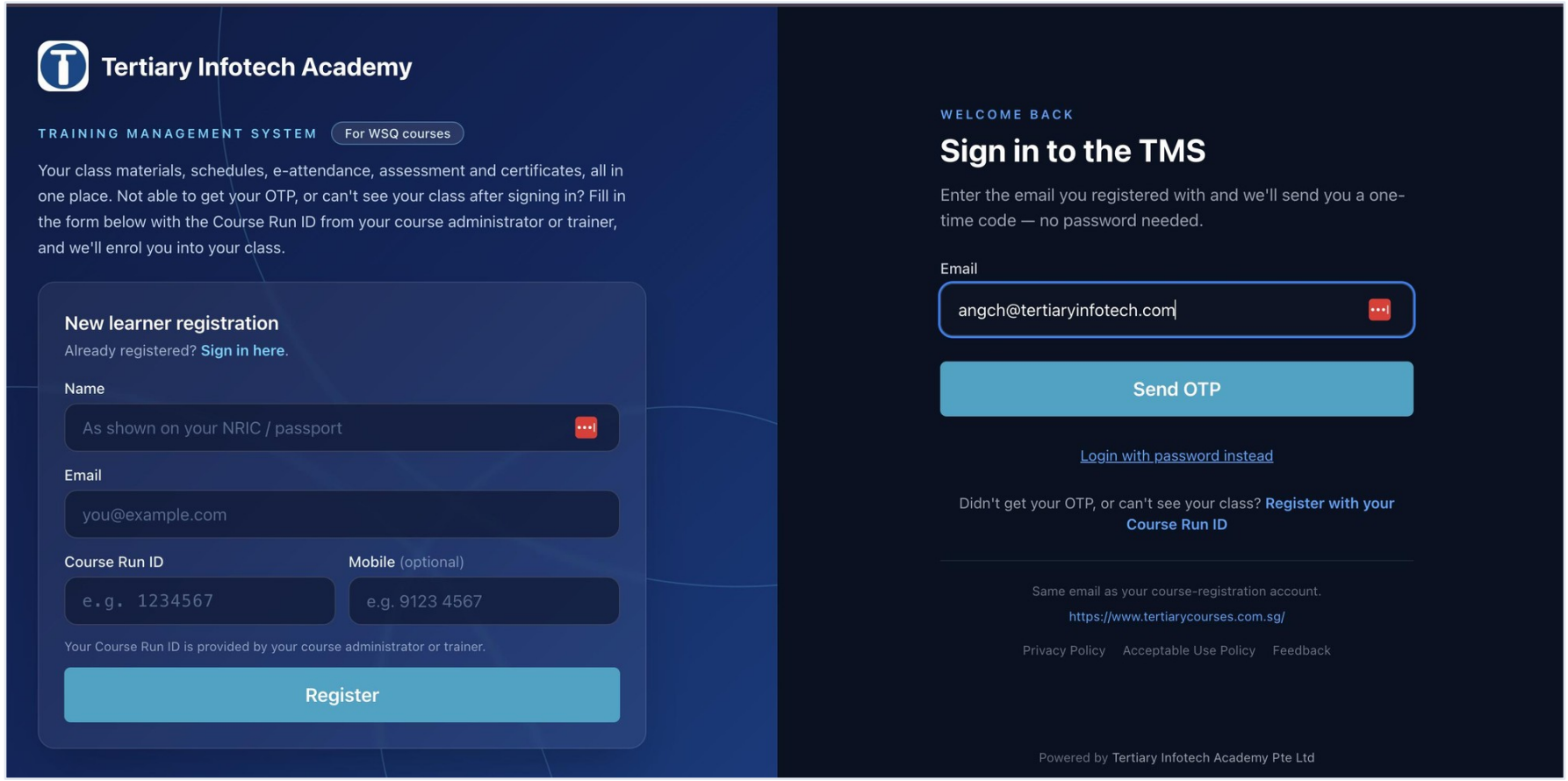
Detailed download and file-access guidance is provided in the Learner Guide.

https://github.com/tertiarycourses/TGS-2024051248-Mastering-Semiconductor-Key-Modules-for-Enhanced-CMOS-Process-Integration-and-Device

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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DOWNLOAD COURSE MATERIAL

Courseware and Assessment on the LMS

1. Download the current learner slides and Learner Guide
2. Use detailed procedures in the Learner Guide and activity folders
3. Complete the WA and PP in the scheduled assessment block
4. Upload completed answers on the LMS

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A fab is a connected system, not a collection of tools.

Every module hands variation, defects and evidence to the next module.

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OFFICIAL TOPIC 1 · LEARNING BLOCK 1A

Patterning, Etch and Fab Control

1A

Build reliable patterns while controlling contamination, CD, overlay and profile.

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Block 1A Roadmap

1 **The Wafer Fab as a System**

The Wafer Fab as a System

2 **Contamination Has Multiple Pathways**

Contamination Has Multiple Pathways

3 **The Lithography Stack**

The Lithography Stack

4 **Mask Tone and Resist Tone**

Mask Tone and Resist Tone

5 **Resist Coating and Bake**

Resist Coating and Bake

6 **Resolution, NA and Process Margin**

Resolution, NA and Process Margin

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Block 1A Roadmap

1 **Overlay, OPC and Mask Protection**

Overlay, OPC and Mask Protection

2 **Wet Etch versus Dry Etch**

Wet Etch versus Dry Etch

3 **Etch Selectivity and Profile**

Etch Selectivity and Profile

4 **Strip and Clean**

Strip and Clean

5 **Pattern-Transfer Window**

Pattern-Transfer Window

6 **Patterning Evidence Gate**

Patterning Evidence Gate

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Patterning, Etch and Fab Control in Practice



1. What module is creating the feature?
2. Which interaction can move device performance?
3. What measurement makes the decision defensible?
4. Where should feedback or containment occur?

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The Wafer Fab as a System

1 **Product flow**

hundreds of tightly coupled operations

2 **Cleanroom**

particles and molecular contamination are controlled

3 **Integration**

each module owns incoming and outgoing evidence

4 **Release logic**

a pass is a measured decision, not an assumption

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Contamination Has Multiple Pathways

1 Particles

block patterns and create opens or shorts

2 Ions

shift electrical behaviour and junction properties

3 Organics

interfere with adhesion, exposure and interfaces

4 Water and tools

transport hidden contamination between modules

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The Lithography Stack

WHAT CHANGES

1. Mask or reticle carries the intended geometry
2. Resist converts optical energy into a soluble contrast

WHAT TO VERIFY

1. Alignment connects the current level to prior levels
2. Developed resist becomes the etch or implant mask

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Mask Tone and Resist Tone

1 **Clear-field and dark-field masks invert where light passes**

Clear-field and dark-field masks invert where light passes

2 **Positive resist removes exposed regions**

Positive resist removes exposed regions

3 **Negative resist retains exposed regions**

Negative resist retains exposed regions

4 **The combination determines whether a hole or island remains**

The combination determines whether a hole or island remains

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Resist Coating and Bake

1 **Prime improves adhesion on the prepared surface**

Prime improves adhesion on the prepared surface

2 **Spin speed and dispense control resist thickness**

Spin speed and dispense control resist thickness

3 **Soft bake removes solvent without damaging image response**

Soft bake removes solvent without damaging image response

4 **Edge bead and coverage affect focus and defectivity**

Edge bead and coverage affect focus and defectivity

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Resolution, NA and Process Margin

Wavelength

shorter light supports smaller features

Numerical aperture

higher NA improves resolution

Depth of focus

normally shrinks as NA rises

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Overlay, OPC and Mask Protection

1 **Overlay error shifts one layer against another**

Overlay error shifts one layer against another

2 **OPC pre-distorts features to counter imaging effects**

OPC pre-distorts features to counter imaging effects

3 **Antireflective layers reduce standing-wave effects**

Antireflective layers reduce standing-wave effects

4 **Pellicles and inspection reduce repeating defects**

Pellicles and inspection reduce repeating defects

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Wet Etch versus Dry Etch

WHAT CHANGES

1. Wet etch is often isotropic and can undercut
2. Plasma/RIE supports directional profiles

WHAT TO VERIFY

1. Chemistry controls material reaction
2. Bias, pressure and transport influence sidewalls and damage

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Etch Selectivity and Profile

1 **Selectivity protects resist and stop layers**

Selectivity protects resist and stop layers

2 **Over-etch clears residual film but consumes margin**

Over-etch clears residual film but consumes margin

3 **Microloading makes local pattern density matter**

Microloading makes local pattern density matter

4 **Endpoint and cross-section evidence localise failure**

Endpoint and cross-section evidence localise failure

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Strip and Clean

1

Wet strip dissolves resist where surfaces tolerate chemistry

Wet strip dissolves resist where surfaces tolerate chemistry

2

Dry strip removes organics but can damage sensitive layers

Dry strip removes organics but can damage sensitive layers

3

RCA-type cleans target particles, organics and metals

RCA-type cleans target particles, organics and metals

4

Drying must avoid watermarking and recontamination

Drying must avoid watermarking and recontamination

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Pattern-Transfer Window

Lithography CD sets the incoming geometry

Lithography CD sets the incoming geometry

Etch bias changes the final feature

Etch bias changes the final feature

Across-wafer signatures expose non-uniformity

Across-wafer signatures expose non-uniformity

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Patterning Evidence Gate

1 **Geometry**

CD, overlay and profile

2 **Defects**

bridge, open, pinhole, residue or collapse

3 **Mechanism**

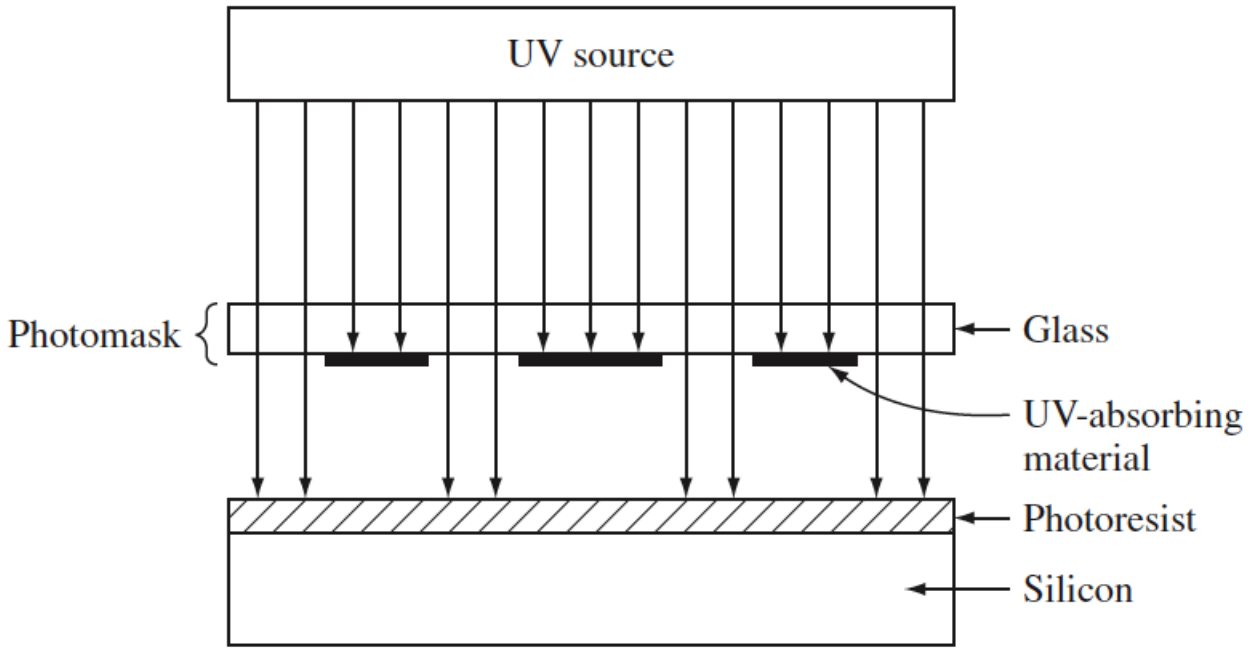
exposure, develop, etch or clean

4 **Action**

release, conditional release, hold or scrap

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Photolithography Exposure Geometry

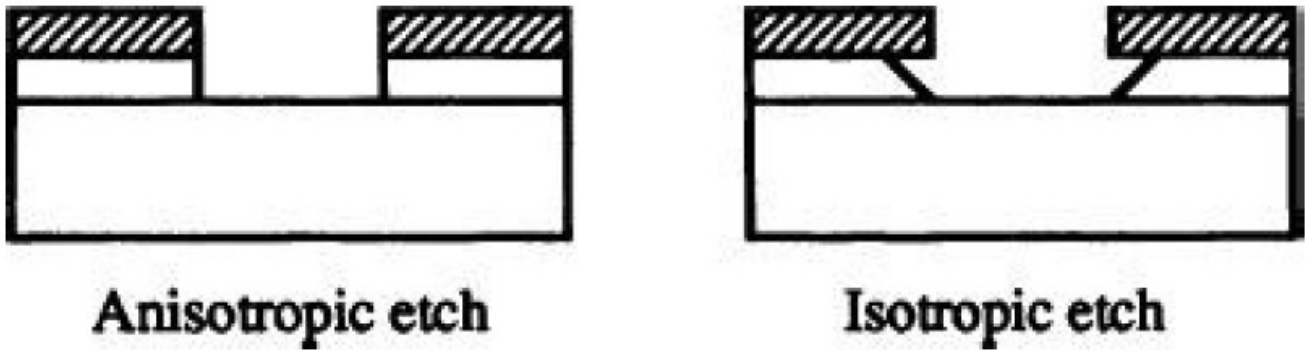


READ THE VISUAL

1. Trace light from source through the mask
2. Connect the aerial image to resist contrast
3. Identify the alignment reference

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Etch Profile and Undercut



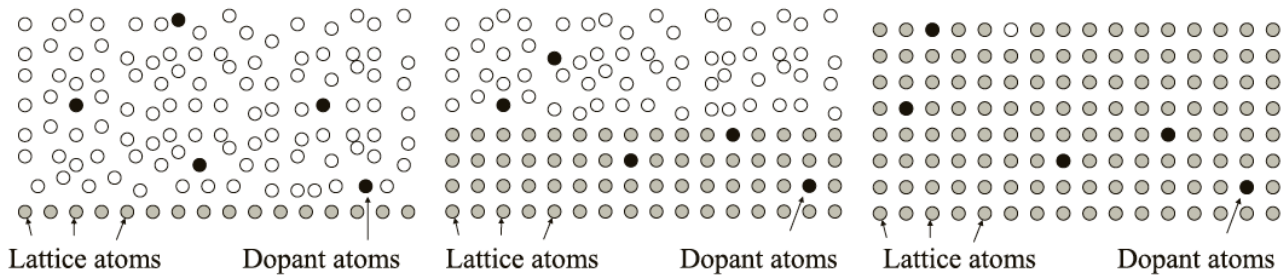
READ THE VISUAL

1. Compare anisotropic and isotropic profiles
2. Locate the material at risk during over-etch
3. Link sidewall shape to selectivity and transport

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LEGACY VISUAL · SOURCE SLIDE 104

Damage Recovery and Dopant Activation

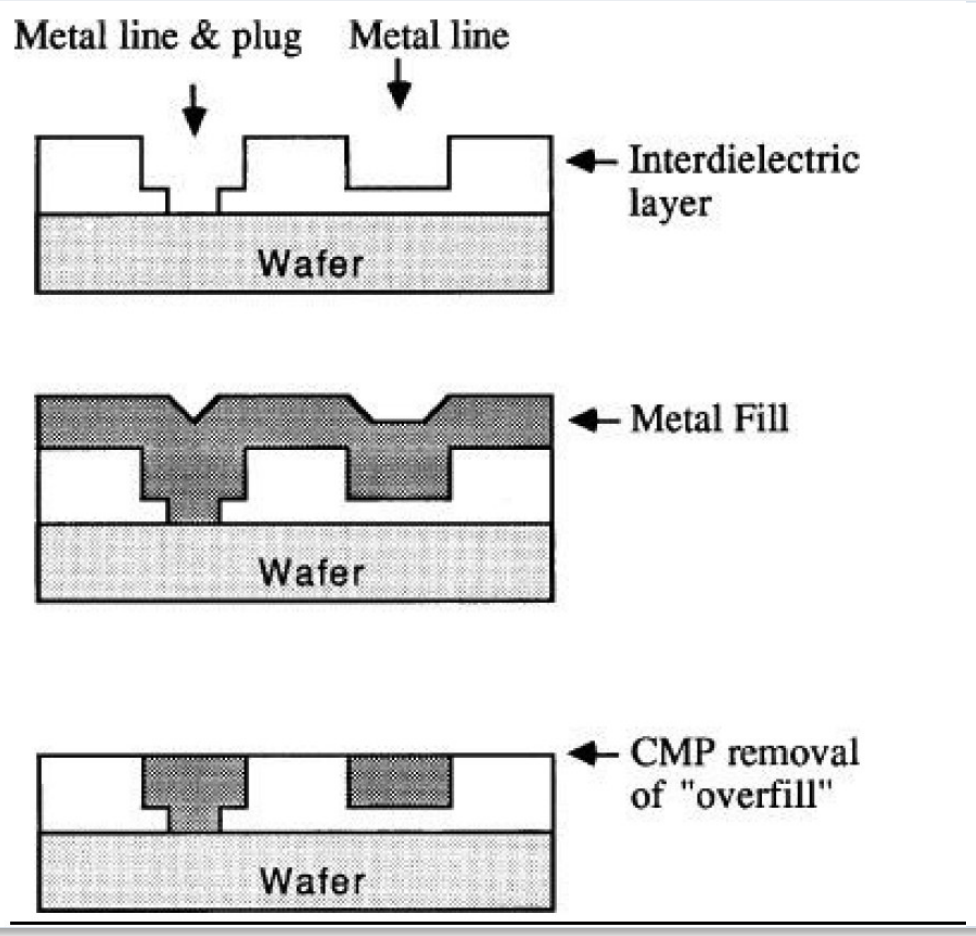


READ THE VISUAL

1. Separate crystal repair from electrical activation
2. Recognise diffusion during the thermal cycle
3. Relate the result to later junction performance

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Dual-Damascene Integration



READ THE VISUAL

1. Follow via and trench formation
2. Locate barrier, seed and copper fill
3. Identify where CMP closes the module

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Build a CMOS Process Module Map

ACTIVITY 1

A new 300 mm logic product is moving from process development into a pilot manufacturing line. The integration team needs one agreed process map before module owners set recipes and control plans.



YOU'LL PRODUCE

Organise the legacy module coverage into a defensible FEOL-to-BEOL flow and expose the interfaces that can propagate variation.

Full procedure: Learner Guide and individual activity folder

LO1 · K1 / A1 / A2 · 45 minutes

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ACTIVITY 1 · ANALYSE BEFORE DECIDING

Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 1 · VERIFY

Acceptance Review

- ✓ **The map contains all seven key process modules and distinguishes FEOL from BEOL.**
- ✓ **At least eight process hand-offs have a measurable characteristic.**
- ✓ **At least three control/feedback loops and two cross-module interactions are justified.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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Diagnose a Pattern-Transfer Process Window

ACTIVITY 2

A contact layer shows intermittent bridging and open defects. Lithography CD, resist profile and plasma-etch selectivity must be reviewed together rather than as isolated module results.



YOU'LL PRODUCE

Connect mask/resist/exposure decisions to etch profile and select a stable corrective action without masking the root cause.

Full procedure: Learner Guide and individual activity folder

LO1 · K1 / A2 · 60 minutes

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ACTIVITY 2 · ANALYSE BEFORE DECIDING

Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 2 · VERIFY

Acceptance Review

- ✓ **Every defect is linked to a plausible lithography/etch mechanism.**
- ✓ **The dominant CD-shift stage is supported by before/after evidence.**
- ✓ **Corrective levers include predicted benefit, risk and verification measurement.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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CHECK YOUR UNDERSTANDING

Block 1A Recap

1 **Explain**

Etch Selectivity and Profile

2 **Explain**

Strip and Clean

3 **Explain**

Pattern-Transfer Window

4 **Explain**

Patterning Evidence Gate

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Lunch Break

45 minutes

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OFFICIAL TOPIC 1 · LEARNING BLOCK 1B

Implant, Thermal, Films and Integrated Flow

1B

Control material modification, film formation, planarity and FEOL-to-BEOL hand-offs.

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Block 1B Roadmap

1. **Implant Dose and Energy**
   Implant Dose and Energy
2. **Beamline Risks**
   Beamline Risks
3. **Crystal Damage and Activation**
   Crystal Damage and Activation
4. **Diffusion, Furnace and RTP**
   Diffusion, Furnace and RTP
5. **Thermal Oxidation**
   Thermal Oxidation
6. **CVD, PECVD and Epitaxy**
   CVD, PECVD and Epitaxy

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Block 1B Roadmap

1 **PVD, Barrier and Seed**

PVD, Barrier and Seed

2 **Copper Dual Damascene**

Copper Dual Damascene

3 **CMP Mechanics and Chemistry**

CMP Mechanics and Chemistry

4 **Planarity Failure Modes**

Planarity Failure Modes

5 **Integrated FEOL and BEOL**

Integrated FEOL and BEOL

6 **Cross-Module Budgeting**

Cross-Module Budgeting

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Implant, Thermal, Films and Integrated Flow in Practice



1. What module is creating the feature?
2. Which interaction can move device performance?
3. What measurement makes the decision defensible?
4. Where should feedback or containment occur?

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Implant Dose and Energy

1 **Dose controls total implanted ions**

Dose controls total implanted ions

2 **Energy controls the depth distribution**

Energy controls the depth distribution

3 **Tilt and twist reduce channeling and shadowing**

Tilt and twist reduce channeling and shadowing

4 **Uniform scanning protects within-wafer consistency**

Uniform scanning protects within-wafer consistency

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Beamline Risks

1. **Charging can deflect beams or damage dielectrics**
   Charging can deflect beams or damage dielectrics

2. **Channeling produces unexpectedly deep tails**
   Channeling produces unexpectedly deep tails

3. **Mask topography creates shadowed regions**
   Mask topography creates shadowed regions

4. **Beam focus and scan strategy affect uniformity**
   Beam focus and scan strategy affect uniformity

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Crystal Damage and Activation

WHAT CHANGES

1. Implant collisions create vacancies and interstitials
2. Anneal repairs lattice damage

WHAT TO VERIFY

1. Activation places dopants on useful sites
2. Electrical improvement can coexist with diffusion

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Diffusion, Furnace and RTP

1

Furnace cycles favour batch throughput and long soak

Furnace cycles favour batch throughput and long soak

2

RTP uses fast ramps and short dwell

RTP uses fast ramps and short dwell

3

Thermal budget controls activation and junction movement

Thermal budget controls activation and junction movement

4

The best cycle balances electrical target with geometry

The best cycle balances electrical target with geometry

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Thermal Oxidation

1

Silicon is consumed as silicon dioxide grows

Silicon is consumed as silicon dioxide grows

2

Dry oxidation grows slowly with strong film quality

Dry oxidation grows slowly with strong film quality

3

Wet oxidation grows faster for thicker films

Wet oxidation grows faster for thicker films

4

Temperature, time and surface condition control integrity

Temperature, time and surface condition control integrity

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CVD, PECVD and Epitaxy

CVD

film forms through surface reactions

PECVD

plasma lowers process temperature

Epitaxy

deposited crystal follows the substrate lattice

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PVD, Barrier and Seed

1 **Sputtering transports target atoms to the wafer**

Sputtering transports target atoms to the wafer

2 **Directional coverage challenges deep features**

Directional coverage challenges deep features

3 **Barrier films limit metal diffusion**

Barrier films limit metal diffusion

4 **Continuous seed enables uniform electrofill**

Continuous seed enables uniform electrofill

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Copper Dual Damascene

WHAT CHANGES

1. Pattern vias and trenches in dielectric
2. Deposit liner, barrier and seed

WHAT TO VERIFY

1. Electrofill copper without seams or voids
2. Polish overburden while protecting pattern density

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CMP Mechanics and Chemistry

1

Pad contact and slurry chemistry work together

Pad contact and slurry chemistry work together

2

Downforce, speed and pad state affect removal

Downforce, speed and pad state affect removal

3

Selectivity protects stop layers

Selectivity protects stop layers

4

Post-CMP clean removes abrasive and metal residue

Post-CMP clean removes abrasive and metal residue

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Planarity Failure Modes

1. **Dishing removes too much metal in wide features**
   Dishing removes too much metal in wide features
2. **Erosion lowers dense pattern regions**
   Erosion lowers dense pattern regions
3. **Incomplete clear leaves conductive residue**
   Incomplete clear leaves conductive residue
4. **Topography and electrical continuity must both pass**
   Topography and electrical continuity must both pass

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Integrated FEOL and BEOL

Isolation and wells define device regions

Isolation and wells define device regions

Gate and source/drain form transistors

Gate and source/drain form transistors

Contacts and interconnect create connectivity

Contacts and interconnect create connectivity

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Cross-Module Budgeting

1. **Thermal budget accumulates across later steps**
   Thermal budget accumulates across later steps

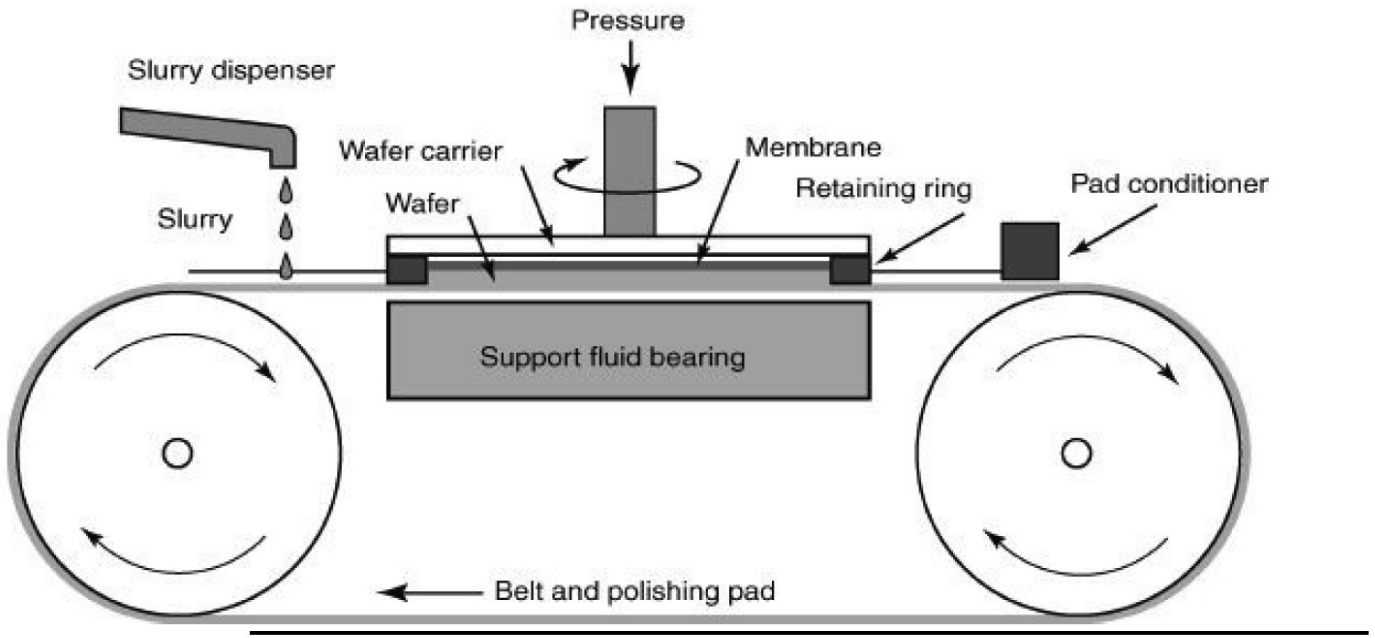
2. **Topography changes the next lithography window**
   Topography changes the next lithography window

3. **Contamination transfers through surfaces and tools**
   Contamination transfers through surfaces and tools

4. **Every hand-off needs evidence and an owner**
   Every hand-off needs evidence and an owner

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CMP Tool and Contact Mechanics



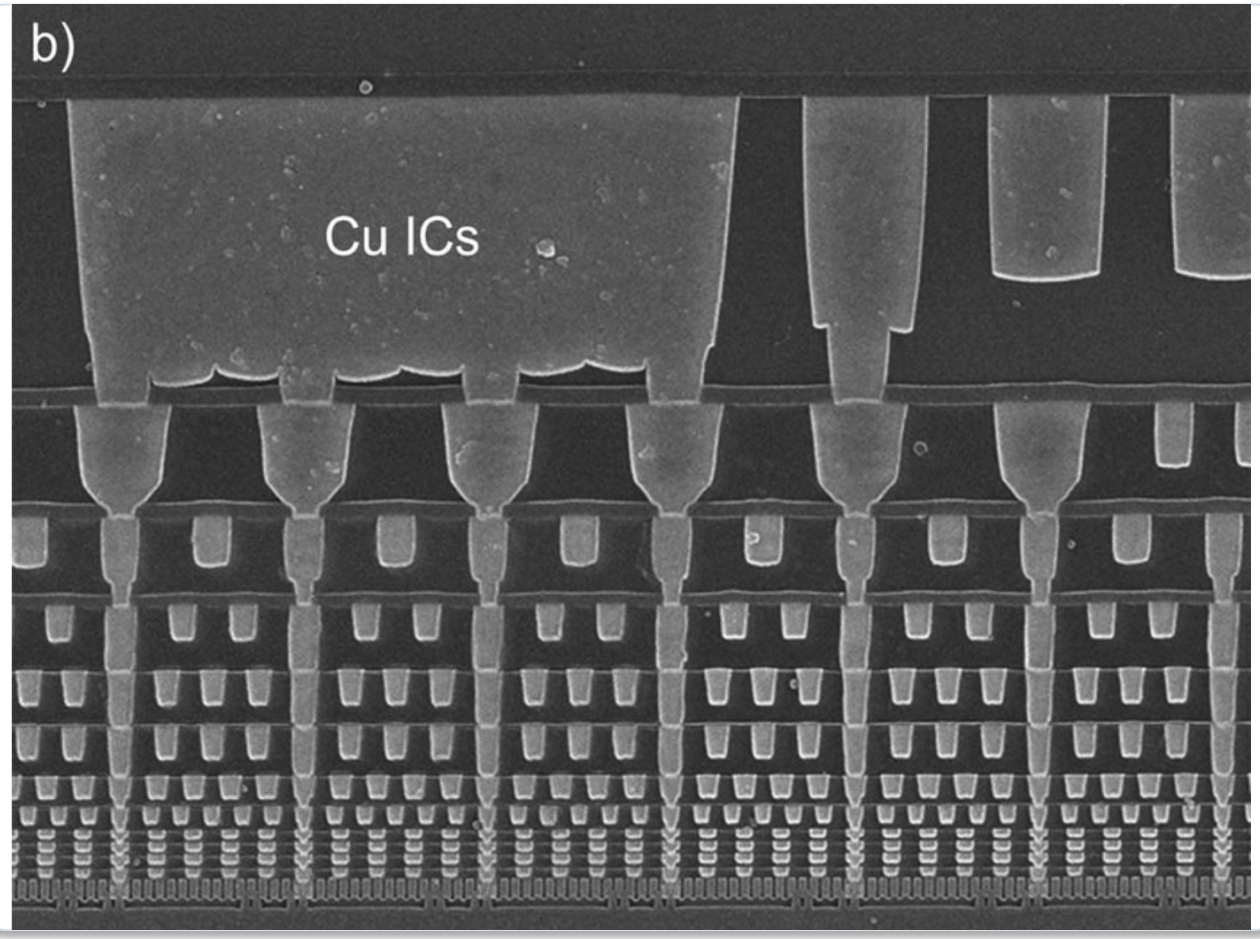
0.32 Chemical–mechanical polishing and planarization

READ THE VISUAL

1. Locate wafer, pad and slurry interface
2. Connect relative motion to removal
3. Distinguish removal rate from final planarity

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FEOL and BEOL Cross-Sections



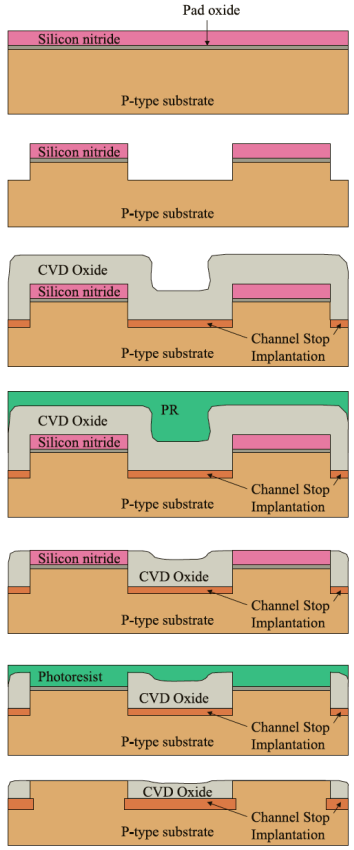
READ THE VISUAL

1. Separate transistor formation from interconnect
2. Read connectivity through contacts and vias
3. Identify module interfaces

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Shallow Trench Isolation Flow

- Clean
- Grow pad oxide
- LPCVD nitride
- STI mask
- Etch nitride and pad oxide
- Strip photoresist
- Etch silicon
- Clean
- Grow barrier oxide
- CVD oxide
- Photoresist coating
- Oxide etchback
- Strip nitride
- Photoresist coating
- Oxide etchback
- Clean
- Oxide annealing

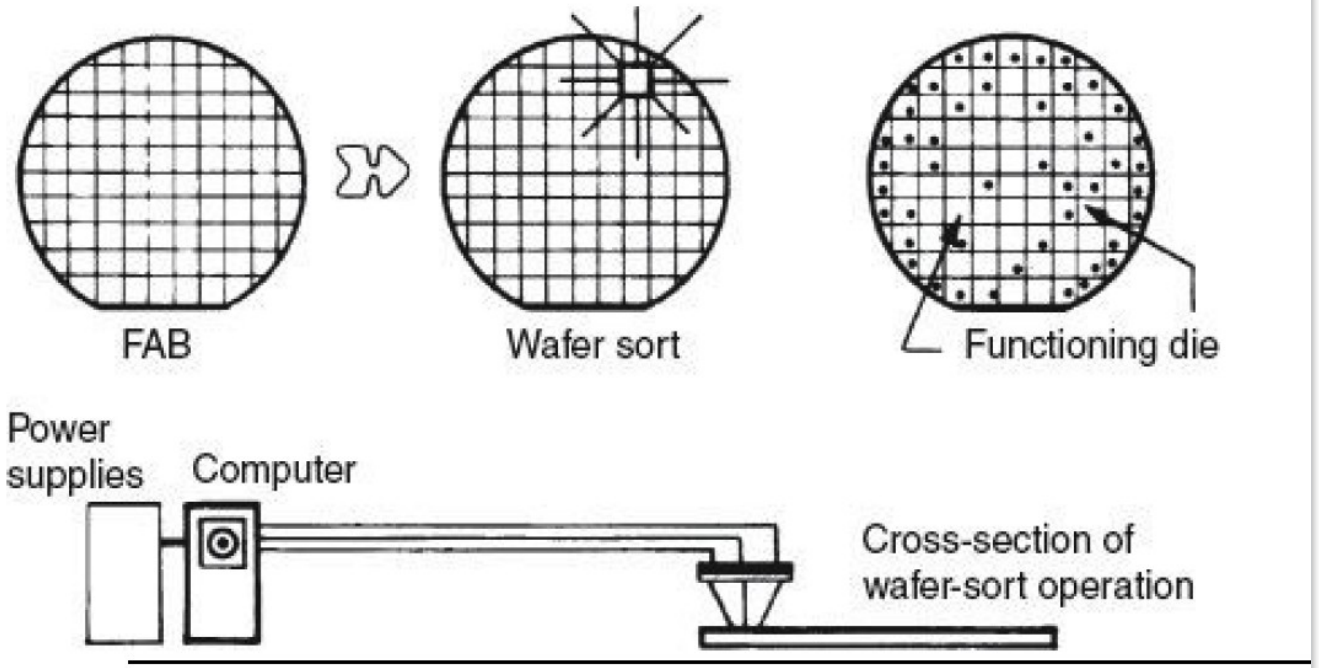


READ THE VISUAL

1. Follow trench from pattern to fill
2. Locate etch, oxidation, deposition and CMP
3. Identify where active-area CD can move

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Wafer Sort as an Integration Gate



READ THE VISUAL

1. Test every die against criteria
2. Convert results into wafer-level patterns
3. Feed spatial signatures back to modules

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Balance Implantation and Thermal Budget

ACTIVITY 3

After a source/drain extension implant, sheet resistance is above target and short-channel control is drifting. The team must distinguish dose/energy error from activation and diffusion effects.



YOU'LL PRODUCE

Create an implant-to-anneal decision trail that balances activation, junction depth, damage recovery and channeling risk.

Full procedure: Learner Guide and individual activity folder

LO1 · K1 / A2 / A3 · 60 minutes

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ACTIVITY 3 · ANALYSE BEFORE DECIDING

Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 3 · VERIFY

Acceptance Review

- ✓ **The recommendation distinguishes implant dose/energy, damage and activation mechanisms.**
- ✓ **Thermal-budget effects on activation and diffusion are explicit.**
- ✓ **Follow-up measurements can confirm or reject the proposed cause.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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Integrate Film Deposition and CMP

ACTIVITY 4

A dual-damascene copper level meets line-resistance targets at wafer centre but fails near the edge with dishing, erosion and suspected barrier continuity loss.



YOU'LL PRODUCE

Trace deposition, fill and CMP interactions and design a verification plan that protects both planarity and electrical performance.

Full procedure: Learner Guide and individual activity folder

LO1 · K1 / A2 / A4 · 60 minutes

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ACTIVITY 4 · ANALYSE BEFORE DECIDING

Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 4 · VERIFY

Acceptance Review

- ✓ **The layer stack and each module's function are correctly labelled.**
- ✓ **The analysis distinguishes deposition/fill causes from polishing causes.**
- ✓ **The release rule uses both physical and electrical evidence.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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CHECK YOUR UNDERSTANDING

Block 1B Recap

1 **Explain**

CMP Mechanics and Chemistry

2 **Explain**

Planarity Failure Modes

3 **Explain**

Integrated FEOL and BEOL

4 **Explain**

Cross-Module Budgeting

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End of Day 1

Return ready to verify performance and control

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OFFICIAL TOPIC 2 · LEARNING BLOCK 2A

Integration Performance, Yield and Reliability

2A

Use metrology, electrical data and reliability evidence to verify product performance.

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Block 2A Roadmap

1 **Isolation and Well Integration**

Isolation and Well Integration

2 **Transistor Formation**

Transistor Formation

3 **Yield Is Multiplicative**

Yield Is Multiplicative

4 **Defect Density and Die Area**

Defect Density and Die Area

5 **Metrology by Question**

Metrology by Question

6 **Measurement Strategy**

Measurement Strategy

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Block 2A Roadmap

1. **Electrical Performance**

   Electrical Performance

2. **Reliability Evidence**

   Reliability Evidence

3. **HCI, TDDB and Electromigration**

   HCI, TDDB and Electromigration

4. **Control Limits versus Specifications**

   Control Limits versus Specifications

5. **Capability**

   Cp and Cpk

6. **Evidence-Based Release**

   Evidence-Based Release

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Integration Performance, Yield and Reliability in Practice



1. What module is creating the feature?
2. Which interaction can move device performance?
3. What measurement makes the decision defensible?
4. Where should feedback or containment occur?

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Isolation and Well Integration

1

LOCOS is simple but creates bird's-beak encroachment

LOCOS is simple but creates bird's-beak encroachment

2

STI supports scaling through etched and filled trenches

STI supports scaling through etched and filled trenches

3

Well architecture controls body regions

Well architecture controls body regions

4

Stress, CD and topography affect device behaviour

Stress, CD and topography affect device behaviour

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Transistor Formation

1 **Gate stack sets electrostatic control**

Gate stack sets electrostatic control

2 **Source/drain implants establish junctions**

Source/drain implants establish junctions

3 **Spacers sequence extension and deep implants**

Spacers sequence extension and deep implants

4 **Contacts must preserve silicide and junction integrity**

Contacts must preserve silicide and junction integrity

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Yield Is Multiplicative

WHAT CHANGES

1. Wafer-fab yield reflects process survival
2. Wafer-sort yield reflects functional die

WHAT TO VERIFY

1. Packaging yield reflects assembly and final test
2. One stage can erase gains made at another

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Defect Density and Die Area

1. **Larger die expose more area to random defects**

   Larger die expose more area to random defects

2. **Clustered defects violate simple random assumptions**

   Clustered defects violate simple random assumptions

3. **Systematic defects need separate treatment**

   Systematic defects need separate treatment

4. **Yield models support prediction, not proof**

   Yield models support prediction, not proof

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Metrology by Question

1 **Ellipsometry**

thin-film thickness and optics

2 **Profilometry/AFM**

step height and topography

3 **SEM/TEM**

CD, profile and internal structure

4 **SIMS/Auger**

concentration and surface composition

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Measurement Strategy

Before and after

isolate the suspected module

Spatial map

expose centre-edge or chamber signatures

Resolution

match instrument to the feature

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Electrical Performance

1 **Four-point probe measures sheet resistance**

Four-point probe measures sheet resistance

2 **Gate-oxide integrity tests dielectric weakness**

Gate-oxide integrity tests dielectric weakness

3 **I-V curves reveal threshold, leakage and drive**

I-V curves reveal threshold, leakage and drive

4 **Product specifications decide pass/fail**

Product specifications decide pass/fail

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Reliability Evidence

WHAT CHANGES

1. Mechanism explains degradation
2. Accelerated stress creates measurable change

WHAT TO VERIFY

1. Failure criterion defines the endpoint
2. Physical confirmation links failure to cause

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HCI, TDDB and Electromigration

1 **Hot carriers damage the channel interface**

Hot carriers damage the channel interface

2 **TDDB accumulates dielectric damage over time**

TDDB accumulates dielectric damage over time

3 **Electromigration moves metal under current density**

Electromigration moves metal under current density

4 **Geometry, material and temperature change lifetime**

Geometry, material and temperature change lifetime

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Control Limits versus Specifications

1 Control limits describe process behaviour

Control limits describe process behaviour

2 Specification limits describe product requirements

Specification limits describe product requirements

3 A stable process may still be off target

A stable process may still be off target

4 An in-spec point can signal a process shift

An in-spec point can signal a process shift

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Capability: Cp and Cpk

Cp compares specification width with spread

Cp compares specification width with spread

Cpk also reflects centering

Cpk also reflects centering

Capability follows stability assessment

Capability follows stability assessment

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Evidence-Based Release

1 **Define the product requirement**

Define the product requirement

2 **Select physical and electrical evidence**

Select physical and electrical evidence

3 **Compare with limits and uncertainty**

Compare with limits and uncertainty

4 **Document disposition and follow-up**

Document disposition and follow-up

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Yield Models and Their Assumptions

Poisson

$$Y = \exp(-A\,D0)$$

Random, independent defects; useful as a simple baseline.

Murphy

$$Y = [(1 - \exp(-A\,D0)) / (A\,D0)]^2$$

Allows defect density to vary across the wafer.

Negative binomial

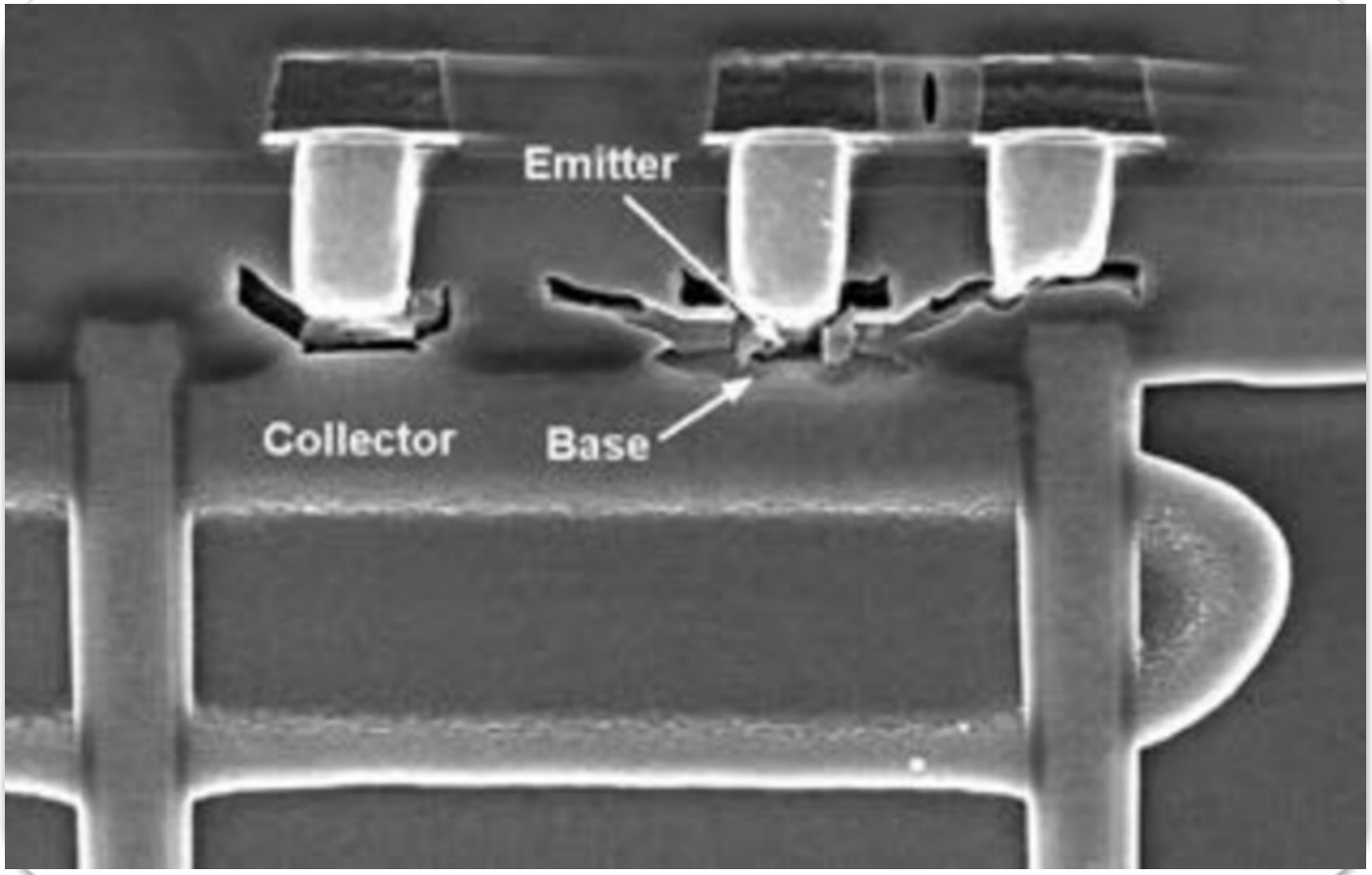
$$Y = (1 + A\,D0 / \alpha)^{(-\alpha)}$$

Represents clustered defects through clustering factor α.

A = die area · D0 = defect density · Use measured yield patterns to test the model assumption.

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SEM Evidence

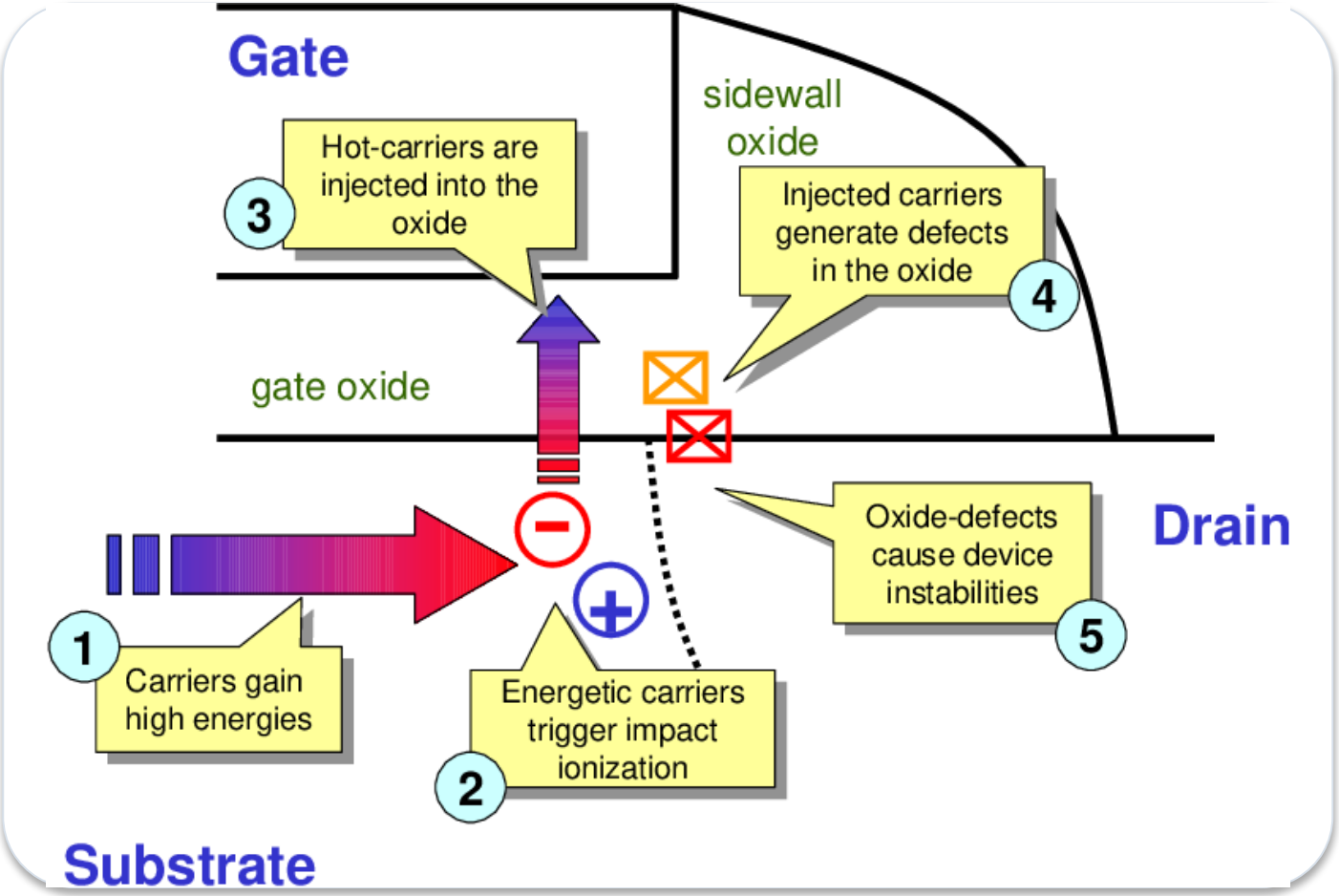


READ THE VISUAL

1. Read dimensions from the scale
2. Distinguish surface from cross-section imaging
3. Link the image to the process question

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Hot-Carrier Degradation

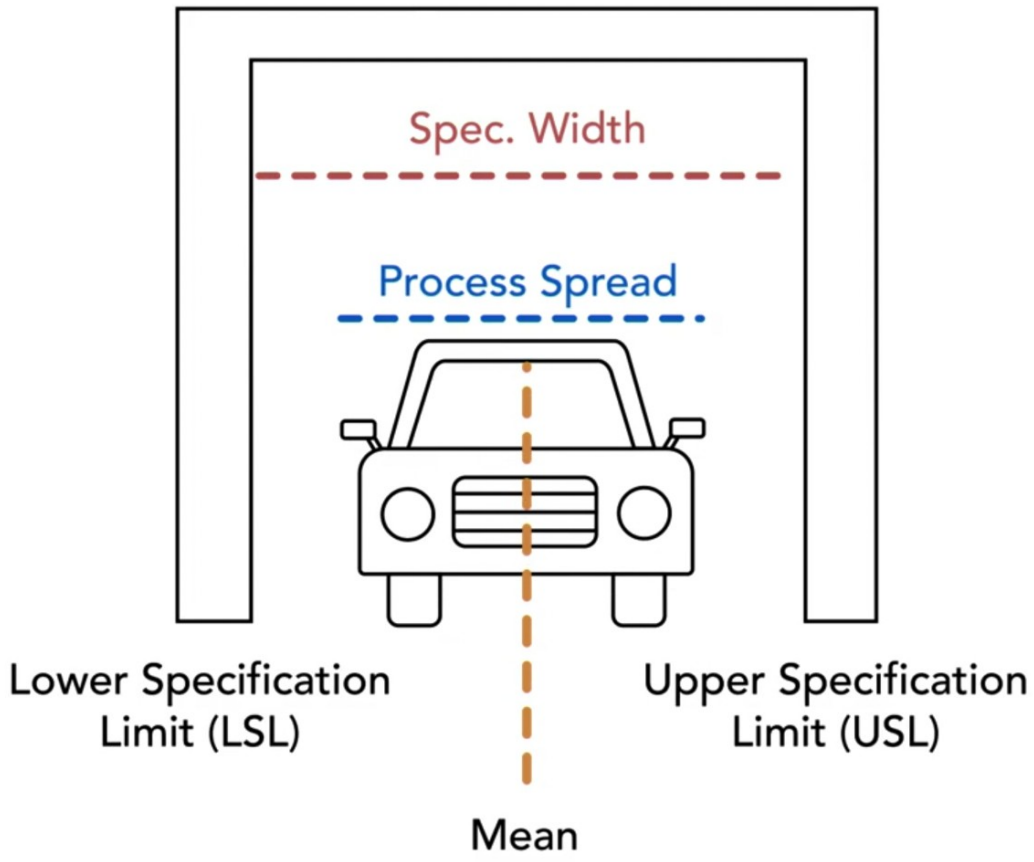


READ THE VISUAL

1. Locate the high-field drain region
2. Connect trapped charge to parameter shift
3. Identify stress and readout evidence

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Capability Is Spread plus Centering



READ THE VISUAL

1. Compare distribution with specifications
2. Separate potential from actual capability
3. Do not claim capability before stability

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Review an STI Integration Flow

ACTIVITY 5

TechSemicon is introducing shallow trench isolation for an advanced CMOS product. Electrical spread and active-area CD variation appear after integration, and the review board needs a complete process/evidence chain.



YOU'LL PRODUCE

Produce the process-flow and challenge analysis used as the foundation for Practical Performance Task 1.

Full procedure: Learner Guide and individual activity folder

LO2 · A1 / A2 / A4 · 75 minutes

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Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 5 · VERIFY

Acceptance Review

- ✓ **The eight core STI stages are in the correct order and have a purpose.**
- ✓ **At least four implementation challenges are linked to measurements.**
- ✓ **The flow explains both scaling benefit and integration risk.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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Analyse Yield and Process Capability

ACTIVITY 6

A gate module has a target CD of 100 nm with specification limits at 90 and 110 nm. The process owner needs Cp/Cpk evidence and a reasoned yield-risk statement before qualification.



YOU'LL PRODUCE

Recreate and improve the legacy ProcessCapability exercise using an auditable Excel workbook with dynamic formulas.

Full procedure: Learner Guide and individual activity folder

LO2 · K2 / A3 / A4 · 75 minutes

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Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 6 · VERIFY

Acceptance Review

- ✓ **Workbook formulas remain intact and show no Excel errors.**
- ✓ **The learner distinguishes Cp from Cpk and links both to process evidence.**
- ✓ **The decision includes containment, corrective action and verification.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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CHECK YOUR UNDERSTANDING

Block 2A Recap

1 **Explain**

HCI, TDDB and Electromigration

2 **Explain**

Control Limits versus Specifications

3 **Explain**

Capability: Cp and Cpk

4 **Explain**

Evidence-Based Release

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Afternoon Break

10 minutes

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OFFICIAL TOPIC 2 · LEARNING BLOCK 2B

SPC, Packaging and Future Integration

2B

Control excursions, close the feedback loop and evaluate next-generation choices.

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Block 2B Roadmap

1 **SPC as a Feedback System**

SPC as a Feedback System

2 **Select the Chart from the Data**

Select the Chart from the Data

3 **Variable Control Charts**

Variable Control Charts

4 **Attribute Control Charts**

Attribute Control Charts

5 **Signals Beyond One Point**

Signals Beyond One Point

6 **Excursion Response**

Excursion Response

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Block 2B Roadmap

1. **Packaging Process Flow**

   Packaging Process Flow

2. **Wire Bond and Flip Chip**

   Wire Bond and Flip Chip

3. **2.5D, 3D and Chiplets**

   2.5D, 3D and Chiplets

4. **FinFET**

   FinFET

5. **GAAFET Nanosheets**

   GAAFET Nanosheets

6. **Beyond Today's CMOS**

   Beyond Today's CMOS

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SPC, Packaging and Future Integration in Practice



1. What module is creating the feature?
2. Which interaction can move device performance?
3. What measurement makes the decision defensible?
4. Where should feedback or containment occur?

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SPC as a Feedback System

1. **Sample the process with a rational subgroup**

Sample the process with a rational subgroup

2. **Plot performance against behaviour-based limits**

Plot performance against behaviour-based limits

3. **Detect special-cause signals**

Detect special-cause signals

4. **Contain risk and restore control with evidence**

Contain risk and restore control with evidence

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Select the Chart from the Data

1 **Variable data**

measurements such as CD

2 **Attribute data**

counts or proportions

3 **Subgroup size decides Xbar-R versus Xbar-S**

Subgroup size decides Xbar-R versus Xbar-S

4 **Opportunity and sample size decide p, np, c or u**

Opportunity and sample size decide p, np, c or u

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Variable Control Charts

WHAT CHANGES

1. Xbar tracks subgroup centre
2. R or S tracks within-subgroup spread

WHAT TO VERIFY

1. The two charts answer different questions
2. A mean chart is incomplete without spread

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Attribute Control Charts

1 **p chart**

proportion defective, variable sample size

2 **np chart**

defective count, constant sample size

3 **c chart**

defect count, constant opportunity

4 **u chart**

defects per unit, variable opportunity

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Signals Beyond One Point

1. **A point beyond a limit is a strong signal**
   A point beyond a limit is a strong signal
2. **Runs and trends expose non-random change**
   Runs and trends expose non-random change
3. **Stratification reveals tool, wafer or time source**
   Stratification reveals tool, wafer or time source
4. **Rules must be defined before interpretation**
   Rules must be defined before interpretation

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Excursion Response

1

Contain

Hold material and preserve evidence

2

Stratify

Tool, chamber, wafer, product and time

3

Correct

Controlled change with rollback criteria

4

Verify

Stable behaviour and capability

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Packaging Process Flow

1. **Prepare and thin wafer**

   Prepare and thin wafer

2. **Separate, inspect and attach die**

   Separate, inspect and attach die

3. **Create electrical interconnect**

   Create electrical interconnect

4. **Seal, plate, final-test and burn-in as required**

   Seal, plate, final-test and burn-in as required

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Wire Bond and Flip Chip

WHAT CHANGES

1. Wire bond is flexible but adds loop length
2. Flip chip raises I/O density

WHAT TO VERIFY

1. Underfill and joint quality manage mechanical risk
2. Inspection and test protect assembly yield

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2.5D, 3D and Chiplets

1

Interposers connect multiple dies

Interposers connect multiple dies

2

3D stacks use vertical connections

3D stacks use vertical connections

3

Known-good-die and thermal design are inputs

Known-good-die and thermal design are inputs

4

Integration control crosses organisations

Integration control crosses organisations

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FinFET

1

A fin raises the channel above substrate

A fin raises the channel above substrate

2

Multiple gate surfaces improve control

Multiple gate surfaces improve control

3

Fin geometry and parasitics affect performance

Fin geometry and parasitics affect performance

4

Patterning and variability challenge manufacturing

Patterning and variability challenge manufacturing

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GAAFET Nanosheets

Gate surrounds the channel

Gate surrounds the channel

Stacked sheets increase effective width

Stacked sheets increase effective width

Release, inner spacer and contacts are critical

Release, inner spacer and contacts are critical

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Technology Readiness Gate

1

Process

Capability and module interactions

2

Performance

Device metrics against target

3

Reliability

Stress evidence and failure criteria

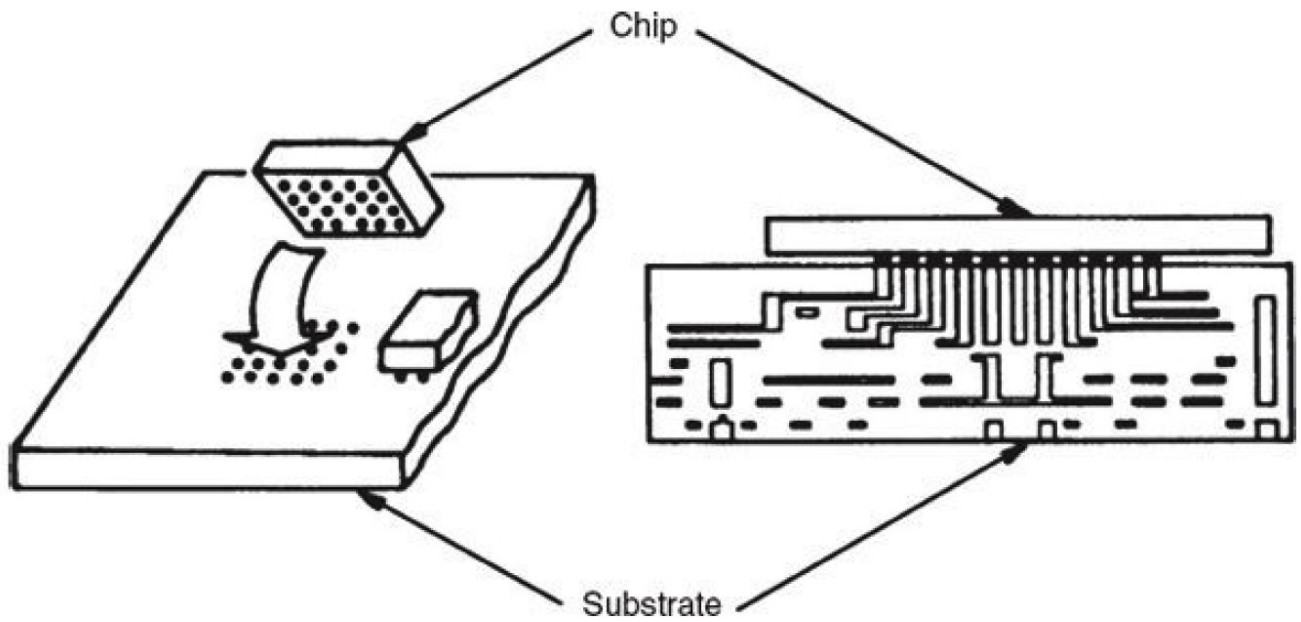
4

Manufacturing

Yield, test and ownership

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Flip-Chip Interconnect

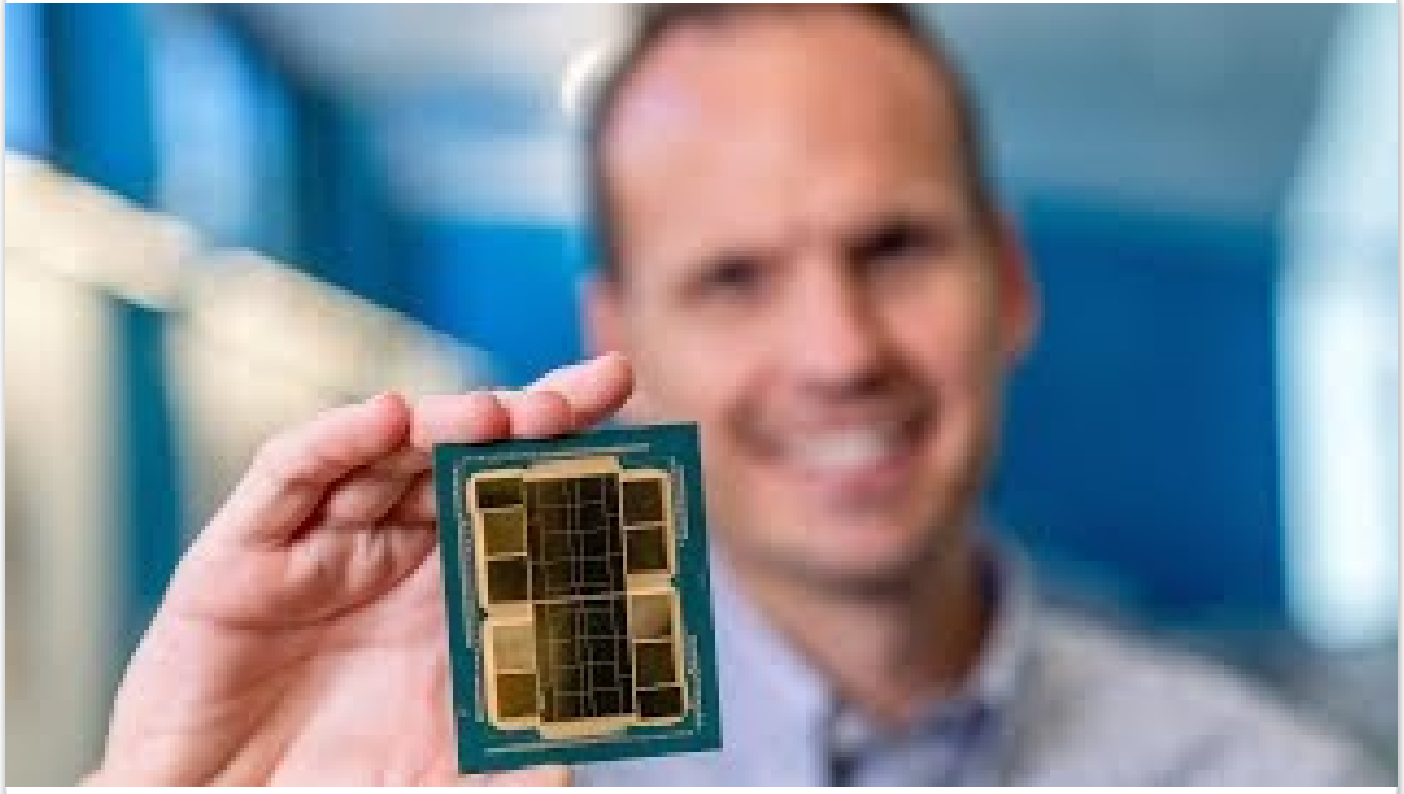


READ THE VISUAL

1. Trace die-to-substrate connections
2. Locate solder joints and underfill
3. Connect geometry to mechanical risk

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Chiplet Integration

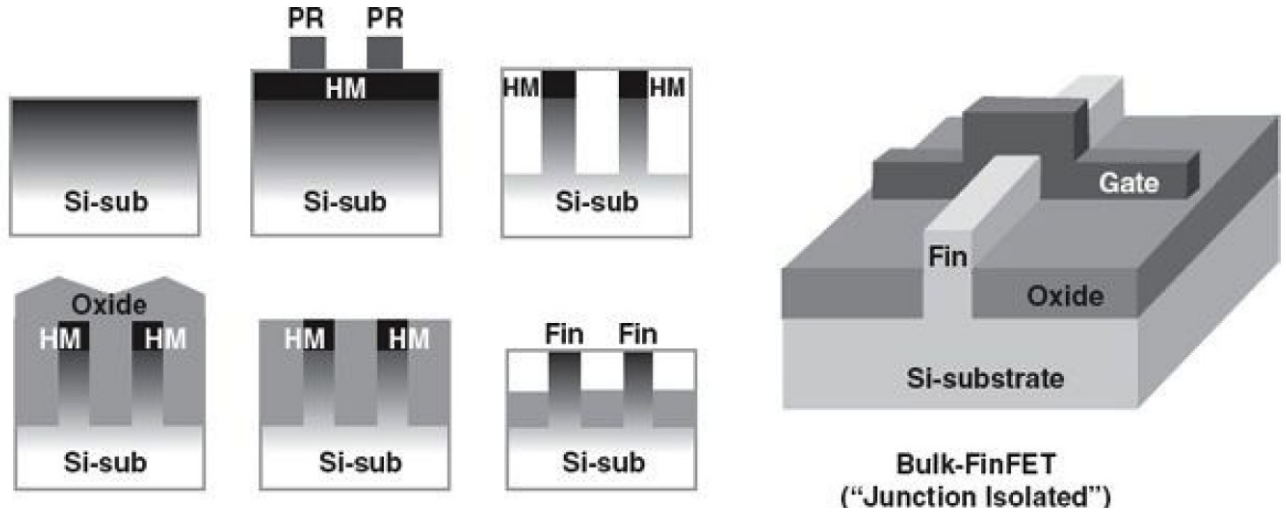


READ THE VISUAL

1. Identify separate functional dies
2. Recognise substrate or interposer
3. Treat known-good-die and test as inputs

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FinFET Geometry

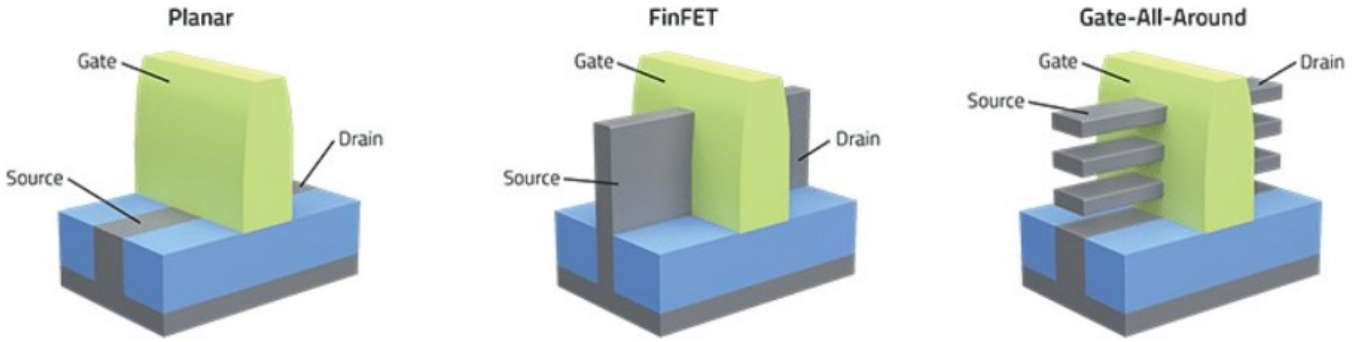


READ THE VISUAL

1. Read gate contact around the fin
2. Connect fin geometry to channel width
3. Identify geometry-sensitive modules

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GAAFET Nanosheet Geometry



READ THE VISUAL

1. See the gate surrounding each sheet
2. Compare stacked channel configurations
3. Identify release and spacer risk

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Respond to an SPC CD Excursion

ACTIVITY 7

STI active-area CD is drifting across production lots. A point crosses the upper control limit and a seven-point upward run appears before the alarm. The team must contain risk and restore control.



YOU'LL PRODUCE

Recreate the legacy P-chart/SPC exercises and generate the corrective-action chain used in Practical Performance Task 2.

Full procedure: Learner Guide and individual activity folder

LO2 · K2 / A3 / A4 / A5 · 90 minutes

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ACTIVITY 7 · ANALYSE BEFORE DECIDING

Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 7 · VERIFY

Acceptance Review

- ✓ **The response identifies both limit violation and non-random pattern.**
- ✓ **Containment precedes root-cause adjustment and protects affected lots.**
- ✓ **Corrective action, verification criteria and follow-up owner are explicit.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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Run a Future-Technology Integration Gate

ACTIVITY 8

A product team is deciding whether to adopt chiplet packaging, FinFET or GAAFET for its next platform. Integration readiness must be assessed using process, performance, reliability and packaging evidence.



YOU'LL PRODUCE

Combine the entire course into a structured technology-gate recommendation without treating future devices as isolated marketing concepts.

Full procedure: Learner Guide and individual activity folder

LO2 · K1 / K2 / A2 / A3 / A4 / A5 · 60 minutes

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ACTIVITY 8 · ANALYSE BEFORE DECIDING

Visual Brief and Evidence Lens



1. Which module interaction is most critical?
2. What evidence distinguishes competing causes?
3. Where is the hold/release gate?
4. What follow-up action closes the risk?

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ACTIVITY 8 · VERIFY

Acceptance Review

- ✓ **The recommendation integrates process, device, reliability, yield and packaging evidence.**
- ✓ **Every gate has a measurable threshold and an accountable follow-up action.**
- ✓ **The conclusion is conditional on evidence rather than technology hype.**

Evidence: analysis output · completed checklist · measurement trail · decision and owner

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CHECK YOUR UNDERSTANDING

Block 2B Recap

1. **Explain**

   2.5D, 3D and Chiplets

2. **Explain**

   FinFET

3. **Explain**

   GAAFET Nanosheets

4. **Explain**

   Beyond Today's CMOS

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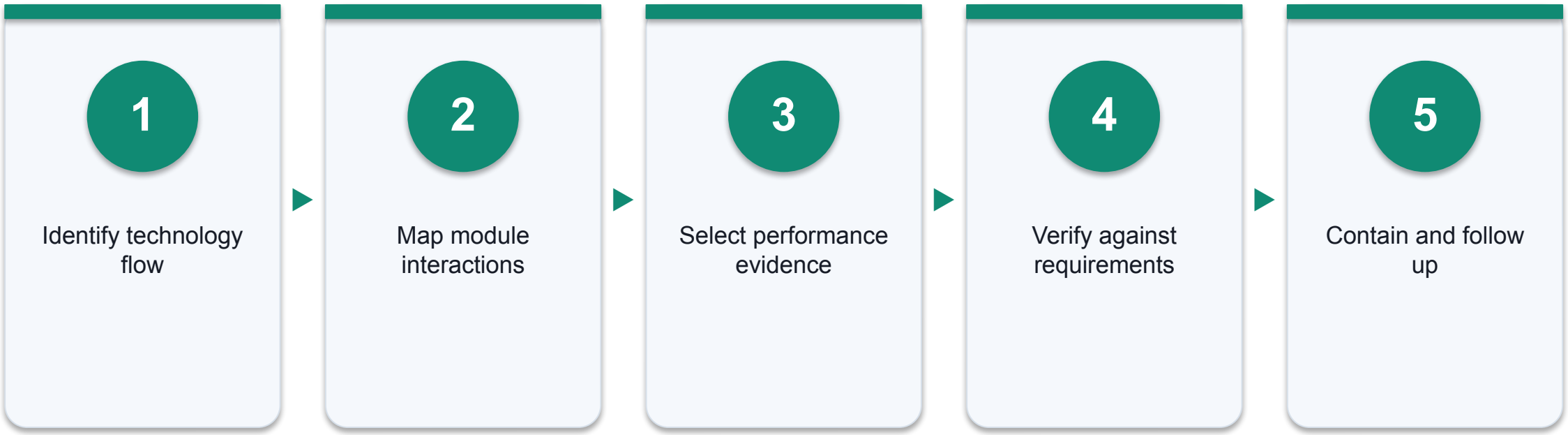
WRAP-UP

Integrated CMOS Decision Workflow

Process · interaction · evidence · performance · action

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The Complete Workflow



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LEARNING OUTCOMES

What You Can Now Do

1

LO1

Identify process flows and modules for specific CMOS technologies in manufacturing.

2

LO2

Determine CMOS device performance and verify process integration in accordance with product specifications.

3

Explain how process interactions propagate

Explain how process interactions propagate

4

Select physical, electrical and statistical evidence

Select physical, electrical and statistical evidence

5

Define a hold/release and follow-up decision

Define a hold/release and follow-up decision

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QUALITY GATE

Professional Integration Review

Process and evidence

1. Is the process sequence complete?
2. Which interaction creates the risk?
3. What incoming and outgoing evidence exists?
4. Are limits and uncertainty explicit?

Decision and ownership

1. Does performance meet product requirements?
2. Is the process stable and capable?
3. Is reliability evidence sufficient?
4. Who owns the next action and by when?

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Source and Evidence Register

1. Legacy v7 trainer deck retained as the content and visual coverage floor
2. Original graphics use aspect-ratio-safe contain placement
3. New Imagegen visuals are recorded in the source register
4. Detailed procedures remain in the Learner Guide and activity folders

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Courseware and Assessment on the LMS

1. Download current learner materials
2. Complete the WA then PP
3. Upload completed answers
4. Contact enquiry@tertiaryinfotech.com for support

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ASSESSMENT

Final Assessment

WA-SAQ

1. Written Assessment
2. 2 open-ended questions
3. K1-K2
4. 60 minutes

PP

1. Practical Performance
2. 2 integrated tasks
3. A1-A5
4. 60 minutes

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ON ASSESSMENT DAY

Assessment Flow



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Digital Attendance (Mandatory)

1. Complete Assessment digital attendance before starting
2. Use the SSG QR code shown by the trainer
3. Submit attendance successfully
4. Ask the trainer if confirmation is not visible

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Thank You

Integrate the flow, follow the evidence and close every risk with an accountable action.

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